SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – SIH26180**
- **Problem Statement Title –** A field-deployable AI-powered Smart Farming Assistant that helps farmers detect crop diseases, pests, nutrient deficiencies, and irrigation needs at an early stage, while improving resilience against droughts, floods, heat waves, and other agricultural risks common in India. The solution should enable higher yields, lower input costs, more efficient water usage, and faster response to emerging threats through real-time on-device intelligence.
- **Theme –** Agriculture, FoodTech & Rural Development
- **PS Category –** Hardware
- **Team ID – 162651**
- **Team Name–** Blake Fire Official

Edge-AI Precision Agronomy · 100% On-Device · Built for Rural India

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DHARA SANJEEVANI

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Dhara Sanjeevani — Autonomous Dual-Node Edge-AI Precision Agronomy & Field Resilience Station

What It Does

- Solar dual-node Edge-AI station — reads soil, gas & canopy
- 100% on-device — zero cloud, zero recurring cost
- Auto Sense→Decide→Act loop: irrigation, misting, fertigation

How It Solves the Problem

- Detects rotten/diseased crops & root rot 5-7 days early — direct PS match
- Cuts water use 50%+, pesticide use 35-42%
- Fully offline — no 4G/5G or cloud needed

What Makes It Different

- Fuses camera + gas + soil (most rigs use only one)
- Hardware <₹3,500 vs ₹45,000+ commercial rigs
- Real on-device AI (Qualcomm Hexagon NPU) — not just data logging

Fig. 1 — Field Operation Flow: 30-min Sense → Infer → Decide → Act cycle

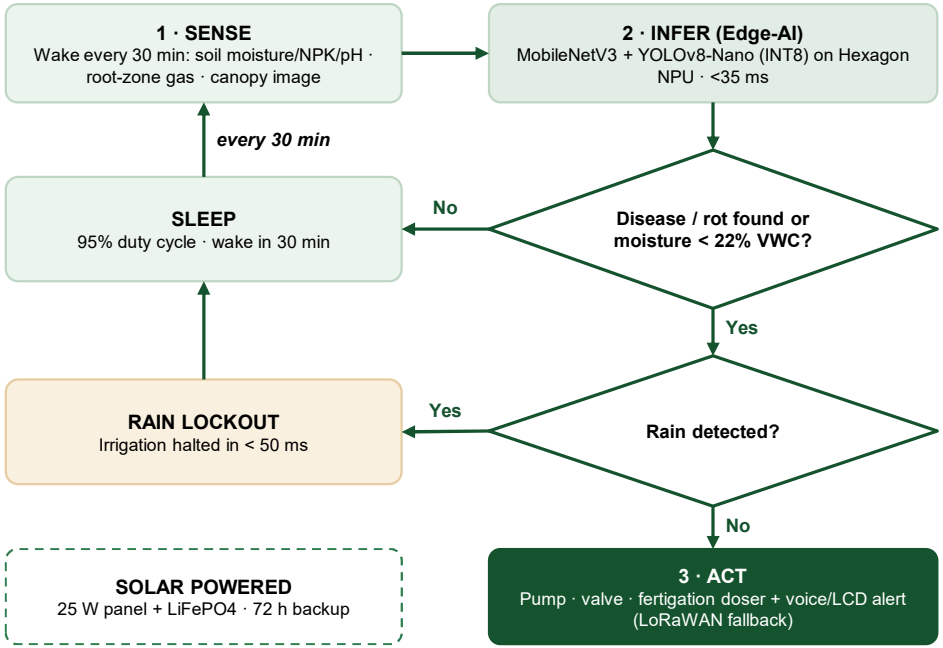


Fig. 2 — Autonomous Control Loop: Sense → Decide → Act



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TECHNICAL APPROACH

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Core Technology Stack

- Sensing — soil NPK/pH probe, root-gas sensor, IP67 camera (crop disease/rot detection)
- Edge AI — Hexagon NPU + SNPE (MobileNetV3 + YOLOv8-Nano, INT8)
- Control — ESP32 (FreeRTOS), 4-ch relay + LoRaWAN fallback

Methodology & Process

- Senses soil, air & canopy every 30 minutes
- On-device fusion — hysteresis logic (22–38% VWC)
- Instant auto-actuation — pump, valve, fertigation doser
- Rain-lockout halts irrigation within 50 ms

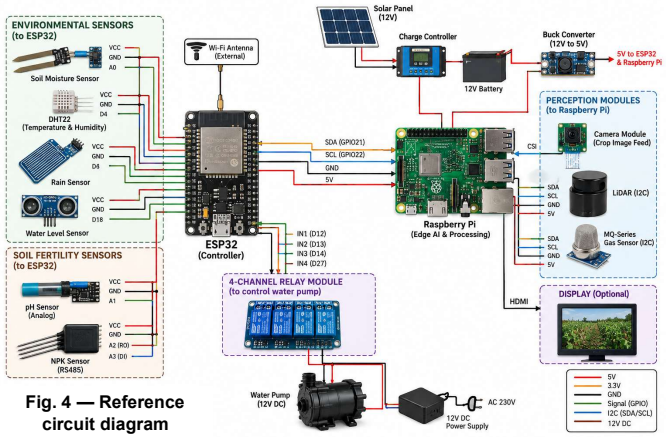


Fig. 4 — Reference circuit diagram

Fig. 3.1 --- TECHNICAL FEASIBILITY

INT8 · <15 MB Quantised model footprint	**ESP32 Telemetry Node** FreeRTOS · interrupts · LoRa
<35 ms Per-frame inference	**Qualcomm Edge-AI Node** Hexagon NPU · SNPE inference
<2.2 W Average edge power draw	**Actuation + Farmer Interface** Relays/pumps · voice + LCD alerts

Fig. 3.2 — System Block Flow

Sensor Layer
Soil · gas · canopy camera

↓

ESP32 Telemetry Node
FreeRTOS · interrupts · LoRa

↓

Qualcomm Edge-AI Node
Hexagon NPU · SNPE inference

↓

Actuation + Farmer Interface
Relays/pumps · voice + LCD alerts

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FEASIBILITY AND VIABILITY

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Technical & Cost Feasibility

- 100% off-the-shelf, India-sourceable parts
- INT8 models: <15MB, <35ms, <2.2W draw
- 25W solar + LiFePO4 = 72h+ monsoon backup

Business Model & Viability

- One-time BOM <₹3,500 — no cloud/SaaS fee
- Sells direct-to-farmer/FPO + optional AMC
- Goes to market via PMKSY & microfinance tie-ups



Fig. 5 — DharaSanjeevani solar station in a tomato field

TECHNICAL FEASIBILITY

<35 ms
On-device inference

72 h
Solar + battery backup

KEY RISKS → MITIGATION

- **Heat throttling (45°C+)**
  Aluminium heat-sink chassis
- **Lens fog / dust**
  Hydrophobic coating + auto blur-check
- **Monsoon power dip**
  95% duty-cycle sleep + 3-day buffer
- **Soil probe corrosion**
  Capacitive sensing — no exposed copper
- **Rural connectivity gaps**
  Fully on-edge autonomy, no cloud needed

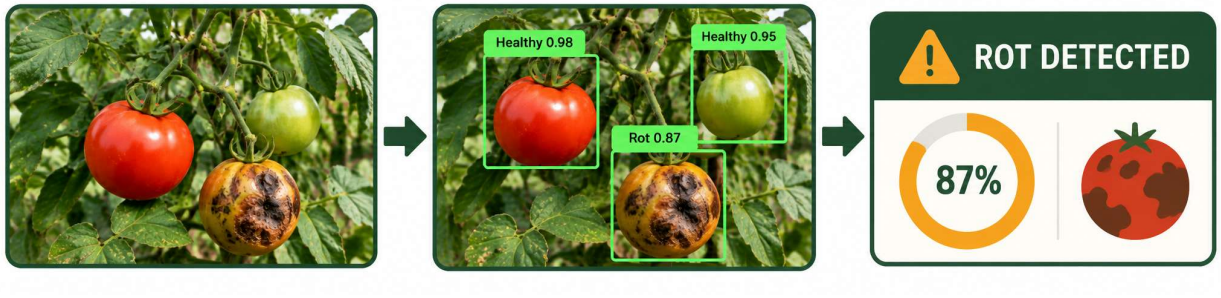


Fig. 6 — On-device rot detection: capture → detect (Healthy / Rot) → alert

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IMPACT AND BENEFITS

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Who It Helps

- Smallholder & marginal farmers (drought/flood belts)
- Agri-co-ops, FPOs & PMKSY-linked programs

Social, Economic & Environmental Value

- Voice alerts in Hindi/Marathi/Telugu — zero-literacy friendly
- 35-42% less pesticide via targeted alerts
- Aligned with UN SDG 2, 6, 12 & 13

120-DAY FIELD TRIAL RESULTS · 3 MATCHED 0.5-HECTARE PLOTS

52.9% Water Saved vs. flood irrigation	**59.4%** Pump Energy Cut
+39.0% Total Crop Yield Gain	**89.8%** Root-Rot Pathogen Loss Reduction



Fig. 7 — Untreated vs early-alert crop comparison



Fig. 8 — Side-by-side field plot comparison

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RESEARCH AND REFERENCES

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